Student Management System: Database Testing Report

Document ID: TER-STUDENT-DB-001 | Target Database: student_qa | Test Execution: Cycle 1

1. Executive Summary & Verification Verdict

This report delivers the verification and functional test results for the Student Management System Database (**student_qa**). A total of 20 database test cases were executed testing CRUD operations, entity constraints, boundary values, referential integrity, and complex analytical SQL queries. All 20 test cases achieved a **100.0% Pass Rate** under strict ANSI/MySQL 8.0+ constraint enforcement. Key findings and configuration requirements (such as MySQL 8.0.16+ requirement for native CHECK constraint execution) were documented.

2. Test Execution Metrics Dashboard

Metric	Count	Percentage	Target Benchmark
Total Database Test Cases Planned	20	100.0%	20 Planned
Total Database Test Cases Executed	20	100.0%	100% Target
Test Cases Passed	20	100.0%	>= 95.0% Goal
Test Cases Failed	0	0.0%	0 Tolerance
Blocked Test Cases	0	0.0%	0 Blockers Target
Documented Risk / Defect Scenarios	4	N/A	Documented in Defect Matrix

3. Category-Wise Performance Breakdown

Category ID & Description	Planned	Pass	Fail	Pass Rate	Core Verification
CAT-01: CRUD & Data Persistence	5	5	0	100.0%	INSERT, SELECT, UPDATE, DELETE & COMMIT
CAT-02: PK, Unique & Not Null	5	5	0	100.0%	Duplicate ID, duplicate email, NULL name blocked
CAT-03: Check Constraints & BVA	4	4	0	100.0%	Age (10, 105) & Marks (-10, 125) rejected
CAT-04: Foreign Key & Referential	3	3	0	100.0%	Invalid FK blocked; RESTRICT delete verified
CAT-05: Analytical SQL Queries	3	3	0	100.0%	INNER/LEFT JOIN, COUNT, duplicate audits

4. Sign-Off & Production Deployment Guidelines

VERDICT: APPROVED / SIGNED OFF. The student_qa relational schema is structurally sound. Production instances must run MySQL 8.0.16+ with strict SQL modes enabled to guarantee continuous enforcement of CHECK constraints and prevent silent precision loss.